

DIAMOND PLANNING

Phase 1 System Design

Demand-driven rough planning, stock assessment and decision support

DOCUMENT PURPOSE This document records only the business rules confirmed during discovery. Manufacturing-stage logic, final rough-plan selection rules and dataset-specific implementation remain open until taught and validated by the business owner.

Version 1.7 - Blue and White Stone Planning Rules

Updated: 17 August 2026

Business owner: Project sponsor

Project assistant: Josh

1. Document Control and Design Status

This controlled Phase 1 baseline captures the agreed interpretation of sales, memo and stock records; category construction; 90-day sales calculations; target-stock rules; and the confirmed rough-planning input, sequencing, grouping, yield and Excel-output rules. Manufacturing-stage calculations and final commercial option selection remain open until taught and validated.

Area	Status	Design position
Sales dataset interpretation	Confirmed	Ready for dataset validation
Memo dataset interpretation	Confirmed	Informational; not part of shortage arithmetic
Finished stock calculation	Confirmed	Lot Status DB = Stock
Category mapping	Confirmed	Lab category + Shape + Weight category
90-day demand calculation	Confirmed	Calculated backward from run date
Manufacturing pipeline	Open	Stages and calculations will be taught later
Rough planning data	Confirmed for output utility	Input, sequencing, grouping and yield rules approved
Final option-selection logic	Open	Planning labels and shortage-based selection remain deferred

1.1 Governance rule

CONTROL No open item in this document is an approved calculation or implementation rule. It must be explained by the business owner, restated by Josh, corrected if needed and explicitly validated before it becomes confirmed design.

1.2 Working method

1. The business owner explains the process, dataset or calculation.
2. Josh restates what was understood and separates facts from assumptions.
3. Questions, edge cases and possible improvements are raised.
4. The business owner validates or corrects the interpretation.
5. Only validated rules are added to the controlled design.

2. Phase 1 Objective and Scope

Phase 1 is focused on improving rough-diamond planning and selection by connecting commercial demand with finished stock, goods already in manufacturing and the possible polished outputs estimated by the planning department.

2.1 Intended business outcome

- Measure recent sales demand consistently at category level.
- Establish the finished-stock target required for each category.
- Show categories where available finished stock is below target.
- Display current memo exposure as supporting information.
- Later incorporate stones already in rough and polished manufacturing stages.
- After pipeline consideration, identify the remaining shortage by category.
- Match planning-department output possibilities against those shortages.
- Highlight commercially useful planning possibilities while retaining yield as an important measure.

2.2 Conceptual decision sequence

6. Read sales, memo and stock data at individual-stone level.
7. Normalize the laboratory name and assign the stone to a weight band.
8. Create the analysis category: Lab category + Shape + Weight category.
9. Calculate the 90-day sales rate and target stock for every category.
10. Compare the target with available finished stock.
11. Show memo pieces separately as commercial context.
12. Later subtract relevant expected manufacturing output from the shortage.
13. Later evaluate rough-planning outputs against any shortage that remains.

3. Source Dataset Specification

The current dataset is stone-level rather than event-history data. Each record represents one diamond stone and carries its latest status.

DATA GRAIN One row = one unique Lot ID = one stone = quantity normally equal to 1.

3.1 Data dictionary

Column	Type/Unit	Confirmed interpretation
Doc Date	Date	Meaning depends on current Lot Status DB.
Lot ID	Identifier	Unique ID for one stone; appears once and does not repeat.
Lot Status DB	Status	Latest/current status; Invoice, Memo and Stock are currently relevant.
Qty	Number	Normally 1 because every row represents one stone.
Shape	Category	Polished-diamond shape used in category mapping.
Weight	Carats	Carat weight of the individual stone.
Color	Grade	Color grade; retained but not used in the current category definition.
Clarity	Grade	Clarity grade; retained but not used in the current category definition.
Lab name	Text	Certification laboratory; blank means Non-Cert.
Certificate	Identifier	Unique certificate number for a certified stone.
Treatment	Text	HPHT, CVD or NULL. NULL means missing data, not untreated.
Sale Total	USD	Final sale value of the individual stone, in US dollars.

3.2 Doc Date interpretation

Lot Status DB	Meaning of Doc Date
Invoice	Date on which the stone was invoiced; this is the sales date.

Lot Status DB	Meaning of Doc Date
Memo	Date on which the stone was issued to the client on memo.
Stock	Date on which the stock was received at the office.

3.3 Latest-status behavior

A Lot ID appears once. If a stone was previously on memo and is later invoiced, its Lot Status DB becomes Invoice. The prior memo status is not retained as a separate row in this dataset.

INTERPRETATION IMPACT Memo records represent the current outstanding memo position, not the complete historical population of stones ever issued on memo.

3.4 Treatment data

Treatment may contain HPHT or CVD. Blank treatment entries must be represented as NULL and reviewed separately through data validation. NULL must never be interpreted as untreated.

4. Category Construction

Every qualifying sales and stock stone is assigned to a common category so that demand and supply can be compared consistently.

CATEGORY KEY Normalized Lab category + Shape + Weight category

4.1 Laboratory normalization

Raw Lab name	Normalized lab category
GIA	GIA
GIA-Premium	GIA
GIA-Standard	GIA
Blank / NULL	Non-Cert
Other values	Rule to be provided when relevant

4.2 Shape

Shape is taken from the Shape column. Shape normalization rules, spelling variants and permitted values will be checked when the example dataset is supplied.

4.3 Weight-band scope

Only stones weighing 1.00 ct and above are in the confirmed scope. The previously listed 0.90-0.99 band is excluded.

Weight category	Lower bound (inclusive)	Upper bound (inclusive)
1.00-1.09	1.00	1.09
1.10-1.49	1.10	1.49
1.50-1.59	1.50	1.59
1.60-1.69	1.60	1.69
1.70-1.99	1.70	1.99
2.00-2.09	2.00	2.09
2.10-2.49	2.10	2.49
2.50-2.59	2.50	2.59
2.60-2.99	2.60	2.99
3.00-3.09	3.00	3.09
3.10-3.49	3.10	3.49
3.50-3.99	3.50	3.99
4.00-4.09	4.00	4.09
4.10-4.49	4.10	4.49
4.50-4.99	4.50	4.99
5.00-5.99	5.00	5.99
6.00-6.99	6.00	6.99
7.00-7.99	7.00	7.99
8.00-8.99	8.00	8.99
9.00-9.99	9.00	9.99
10.00-14.99	10.00	14.99
15.00-19.99	15.00	19.99
20.00-24.99	20.00	24.99
25+	25.00	No upper limit

4.4 Category example

Input	Value
Lab name	GIA-Premium
Shape	Oval
Weight	1.07 ct
Final category	GIA + Oval + 1.00-1.09

5. Sales Analysis and Target-Stock Calculation

5.1 Sales eligibility

A record qualifies for the sales calculation only when Lot Status DB equals Invoice and Doc Date falls within the rolling 90-day window calculated backward from the analysis run date.

DATE WINDOW Use the current run date as the endpoint. Include today and the preceding 89 calendar dates to form an inclusive 90-day window.

5.2 Calculation specification

Measure	Confirmed calculation
90-day sales pieces	Count qualifying Invoice stones in the category.
Monthly average	90-day sales pieces / 3.
Unrounded target stock	Monthly average x 2.
Target stock	Conventionally round unrounded target to the nearest whole piece; exact .5 rounds upward.
Available stock	Count stones in the same category where Lot Status DB = Stock.
Initial shortage	Maximum of 0 and (Target stock - Available stock).

5.3 Conventional rounding

Target stock is held in whole pieces. Conventional rounding is used rather than ceiling rounding or banker's rounding.

Unrounded value	Rounded target
5.33	5 pcs
5.50	6 pcs
5.67	6 pcs

5.4 Worked example

Measure	Value
Category	GIA + Round + 1.00-1.09

Measure	Value
Invoice pieces in last 90 days	10
Monthly average	$10 / 3 = 3.33$
Unrounded target stock	$3.33 \times 2 = 6.67$
Rounded target stock	7 pcs
Available Stock pieces	4 pcs
Initial shortage	$\max(0, 7 - 4) = 3$ pcs

5.5 Required category output

At minimum, the analysis output should expose the following fields so every shortage is explainable:

- Normalized Lab category, Shape and Weight category.
- 90-day sales pieces.
- Monthly sales average.
- Unrounded target stock and rounded target stock.
- Available finished-stock pieces.
- Initial shortage pieces and shortage indicator.
- Current memo pieces as a separate informational measure.
- Manufacturing-pipeline contribution and final pipeline-adjusted shortage when those rules are confirmed.

6. Stock and Memo Interpretation

6.1 Available finished stock

Available stock is calculated only from records whose Lot Status DB equals Stock. The count is performed in pieces within the same normalized Lab-Shape-Weight category used for sales.

6.2 Memo stock

Memo stock is not included in the current required-stock or shortage arithmetic. It is shown separately to indicate how many goods are currently issued to clients on credit/memo.

MEMO RULE Memo is decision context, not an arithmetic adjustment to the current stock requirement.

6.3 Shortage indication

A category must be marked short when available finished-stock pieces are lower than the rounded target-stock quantity.

Condition	Output
Available stock < Target stock	Short = Yes; show positive shortage quantity
Available stock = Target stock	Short = No; shortage = 0
Available stock > Target stock	Short = No; shortage = 0

7. Manufacturing Pipeline - Conceptual Design

Current manufacturing includes stones at different stages of rough working and different stages of polished-diamond working. The detailed stages, subtasks, confidence rules and calculations have not yet been provided and must not be inferred.

7.1 Confirmed purpose

Pipeline goods will later be considered after finished stock and before the system passes a required shortage to rough-plan selection. This prevents recommending production for categories that are already expected from work in process.

7.2 Future calculation position

CONCEPT ONLY Final pipeline shortage = Target stock - Available finished stock - relevant expected manufacturing contribution. The definition of 'relevant expected manufacturing contribution' remains open.

7.3 Topics awaiting training

- Rough-processing stages and their subtasks.
- Polished-processing stages and their subtasks.
- Which stages have reliable expected shape and weight information.
- How expected pieces are mapped to categories.
- Whether and how stage uncertainty affects the quantity counted.
- Treatment of losses, re-planning, delays, rejection and duplicate pipeline records.

8. Rough Planning - Confirmed Concepts

A rough diamond may have many possible plans, and the form of those possibilities can vary from rough to rough. Some possibilities may produce one large main stone; others may produce multiple smaller stones or a combination of shapes and weights based on the planner's estimation.

8.1 Illustrative planning possibilities

For a 5.00 ct rough diamond, the planning department might estimate possibilities such as one Round of 3.50 ct, one Oval of 4.00 ct, one Emerald of 4.10 ct, or a multi-piece combination. These examples explain variability; they are not an approved option-selection rule.

8.2 Yield formula

CONFIRMED FORMULA Yield % = (Total expected polished weight of all pieces in the planning possibility / Original rough weight) x 100

For a multi-piece possibility, the expected weights of all pieces are summed before dividing by the original rough weight.

8.3 Future category matching

Expected polished pieces will ultimately be mapped using the same normalized Lab-Shape-Weight category structure so their usefulness against remaining shortages can be assessed. The raw planning workbook structure, Stone Name parsing, output layout, plan sequencing and yield grouping rules are now confirmed in Section 15. Planning labels and the final selection decision remain open.

8.4 Deliberately unresolved decision

REMINDER REQUIRED When a high-yield possibility produces an unrequired category but a lower-yield possibility produces a required category, the selection rule is not yet decided. Revisit this only after datasets, pipeline logic and real business cases are aligned.

9. End-to-End Phase 1 Workflow

14. Load stone-level sales, memo and stock records.
15. Validate Lot ID uniqueness, Qty, dates, weights, statuses and critical category fields.
16. Represent missing Treatment values as NULL and report them through data validation.
17. Normalize Lab name, including grouping GIA, GIA-Premium and GIA-Standard as GIA; map blank Lab name to Non-Cert.
18. Exclude stones below 1.00 ct from the confirmed category analysis.
19. Assign each qualifying stone to a weight band and construct Lab + Shape + Weight category.
20. Count Invoice pieces in the rolling 90-day window.
21. Calculate monthly average, unrounded target and conventionally rounded target stock.
22. Count available Stock pieces and compute the initial shortage.
23. Display current Memo pieces separately for context.
24. Later incorporate validated manufacturing-stage contributions.
25. Pass only the remaining pipeline-adjusted shortage to rough-planning evaluation.
26. Later compare planning possibilities, yield and commercial usefulness using rules confirmed from example data.

10. Data Validation Requirements

The following controls are recommended to protect calculation quality. They are validation checks and do not change confirmed business calculations.

Check	Expected condition	Suggested handling
Lot ID uniqueness	No duplicates	Flag duplicates; exclude from final reporting until resolved
Quantity	Normally 1	Flag NULL, zero, negative or values other than 1
Weight	Numeric and positive	Flag missing, non-numeric or non-positive values
Doc Date	Valid date	Flag missing, invalid or future dates
Lot Status DB	Recognized value	Use Invoice/Memo/Stock for current analysis; report others
Shape	Present and normalized	Flag blank or spelling variants for business review
Lab name	Known value or blank	Map blank to Non-Cert; review unexpected names
Treatment	HPHT, CVD or NULL	Keep NULL; report missingness separately
Certificate	Unique where populated	Flag duplicates and inconsistent lab/certificate combinations
Sale Total	Numeric USD for relevant sales	Flag missing, negative or non-numeric sales values
Weight band	Mapped for weights ≥ 1.00	Flag any in-scope weight that fails to map

10.1 Suggested reconciliation

- Reconcile the total qualifying Invoice rows before and after category mapping.
- Reconcile the total qualifying Stock rows before and after category mapping.
- Show excluded rows by reason, including weight below 1.00 ct, unsupported status and invalid fields.
- Retain both raw and normalized values so every transformation can be traced.

11. Initial Output Specification

The first analytical output should be a category-level result with transparent intermediate calculations rather than only a final shortage flag.

Output field	Purpose
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Output field	Purpose
Lab category	Normalized certification group
Shape	Commercial shape dimension
Weight category	Confirmed weight band
90-day sales pcs	Recent invoiced demand count
Monthly average pcs	90-day count divided by 3
Unrounded target pcs	Monthly average multiplied by 2
Target stock pcs	Conventionally rounded whole-piece target
Available Stock pcs	Current finished stock
Initial shortage pcs	Target minus Stock, floored at zero
Short indicator	Yes when available Stock is below target
Current Memo pcs	Separate informational exposure
Manufacturing contribution	Future field; rule not yet confirmed
Final shortage pcs	Future pipeline-adjusted result

11.1 Recommended visibility

Josh's recommendation is to preserve the intermediate figures in the user interface and exports. Showing only a red/green shortage signal would make the result difficult to validate and could hide data-quality issues.

12. Open Decisions and Future Training Register

ID	Open topic	Required next input
O-01	Other Lab name normalization	Actual values and grouping rules
O-02	Shape normalization	Example dataset and approved shape list
O-03	Manufacturing stages	Rough and polished subtask definitions
O-04	Pipeline quantity calculation	Stage-level eligibility and certainty rules
O-05	Planning dataset structure - confirmed	Fixed 11-column input contract documented in Section 15
O-06	Planning possibility grouping - confirmed	First 17 individual; later rows grouped by upward Est. Weight reset

ID	Open topic	Required next input
O-07	Final option selection	Rules balancing yield and shortage usefulness
O-08	Expected demand beyond 90-day sales	Forecasting method when introduced
O-09	Other Lot Status DB values	Definitions if they later enter scope
O-10	Sales/memo/stock edge cases	Returns, cancellations, status timing and data corrections if applicable

12.1 Mandatory future reminder

Before finalizing rough-plan recommendation logic, explicitly revisit the case where maximum yield conflicts with production of currently required stock.

13. Acceptance Rules for the Confirmed Sales-and-Stock Module

- Every Lot ID is treated as one stone and counted once.
- Only Invoice records in the rolling 90-day window contribute to sales pieces.
- The 90-day window is based on the actual run date.
- GIA, GIA-Premium and GIA-Standard all map to GIA.
- Blank Lab name maps to Non-Cert.
- Blank Treatment remains NULL and is not treated as untreated.
- Only confirmed weight bands beginning at 1.00 ct are analyzed.
- The monthly average equals 90-day sales pieces divided by 3.
- Target stock equals two times monthly average, conventionally rounded; .5 rounds upward.
- Available stock counts only Lot Status DB = Stock.
- Shortage is positive only when rounded target exceeds available Stock pieces.
- Memo is shown separately and does not adjust the shortage calculation.
- Manufacturing-stage and final commercial option-selection logic remain pending; the confirmed planning workbook transformation and yield rules in Section 15 may be used operationally.

14. Change Log

Version	Date	Description	Status
1.0	11 Aug 2026	Initial system-design baseline created from validated discovery conversation	Controlled draft
1.1	17 Aug 2026	Added fixed planning-input schema, Stone Name parsing, source-order preservation, first-17 individual yield,	Controlled draft

Version	Date	Description	Status
		additional-plan grouping, merged Pcs/Yield cells and utility output rules	
1.2	17 Aug 2026	Additional-group boundaries now compare conventionally rounded two-decimal Est. Weight values; output weights and Yield % retain original precision	Controlled draft
1.3	17 Aug 2026	Added Sarin-to-ecosystem Shape normalization, including Ratio-based classification of EMERALD 5STEP	Controlled draft
1.4	17 Aug 2026	Restricted EMERALD 5STEP Asscher classification to Ratio 1.00-1.03; invalid gap/out-of-range values are highlighted and block export	Controlled draft
1.5	17 Aug 2026	Set all output Est. Weight values to a three-decimal display while retaining two-decimal comparison for additional-group boundaries	Controlled draft
1.6	17 Aug 2026	Repeat Packet on every planning row and remove blank separator rows between Stone Name blocks	Controlled draft
1.7	17 Aug 2026	Added Blue/White stone modes, White Stone variable-length Packet parsing, 17/32 main-	Controlled draft

Version	Date	Description	Status
		plan limits, fixed additional-group pastel highlighting and top-three Yield % ranking highlights	

NEXT CONTROLLED STEP Review this baseline with the business owner. Corrections should be validated before the document is revised. The next functional training area can then be introduced with example datasets.

15. Planning Output Utility - Confirmed Rules

This section is the controlled specification for converting the planning department's raw Excel workbook into the structured planning output. These rules apply independently within each Stone Name.

15.1 Input workbook contract

- The input filename is variable and must never be used as a processing rule.
- The utility reads the first worksheet of an .xlsx workbook.
- Future input workbooks contain data without a header row. A legacy header row may be detected and skipped automatically.
- Only the first 11 columns are interpreted. Their physical sequence is fixed.

Position	Raw field	Output meaning
1	[STONE.Name]	Stone Name and identity parsing
2	[RESULT.1.Rough Cut]	Rough Weight
3	[RESULT.1.ShapeName]	Shape
4	[Result.1.PolishWeight]	Est. Weight
5	[RESULT.CUR.Clarity]	Clarity
6	[RESULT.CUR.COLOR]	Color
7	[RESULT.CUR.TotalDepth_%]	Depth %
8	[RESULT.CUR.Ratio]	Ratio
9	[RESULT.CUR.Length]	Length
10	[RESULT.CUR.Width]	Width
11	[RESULT.CUR.TotalDepth_mm]	MM

15.2 Stone Name parsing and row identity

Blue Stone format example: 670D-764_E+pv.

- Kapan is everything before the first hyphen: 670D.
- Packet is the value between the hyphen and underscore: 764.
- Signer Name is everything after the plus sign: pv. Original capitalization is preserved.
- The meaning of _E is unresolved and no business logic is applied to it.
- White Stone format example: 2501-001 HA. Kapan is everything before the hyphen (2501). Packet is the complete numeric value between the hyphen and the following space (001). Packet length is variable, so both 001 and 1 are valid, and leading zeros must be preserved. Signer Name is the rightmost value after the space (HA). All identical Stone Name rows form one planning block. Rough Weight must remain consistent throughout that block.

15.3 Source-order preservation

Planning rows must be written in exactly the sequence received. The utility must not sort or reorder plans by Est. Weight, Shape or any other field. Stone Name blocks retain their order of first appearance. Different Stone Name blocks must follow one another continuously; no blank separator row is inserted between or inside blocks.

15.4 Output-column sequence

Col.	Header	Confirmed use
A	NO.	Plan number within the Stone Name; starts at 1
B	DATE	Planning date; first row of the Stone Name only
C	SIGNER NAME	Parsed signer; first row only
D	NO.	Sequential Stone Name/block number; first row only
E	KAPAN	Parsed Kapan; first row only
F	Packet	Parsed Packet; repeated on every planning row within the Stone Name
G	Rough Weight	Repeated on every plan row
H	Blank label column	Blank for main plans; merged N Pcs label for additional groups
I	Shape	Raw planning shape
J	Est. Weight	Expected polished weight; displayed to exactly three decimal places
K	Clarity	Estimated clarity
L	Color	Estimated color
M	Depth %	Raw total depth percentage; not Yield %
N	Ratio	Raw ratio
O	Width	Raw width
P	Length	Raw length
Q	MM	Raw total depth in millimetres

Col.	Header	Confirmed use
R	Yield %	Individual or merged group yield; displayed to two decimal places and ranked within each Stone Name

15.5 Main plans by stone type

Plan numbering restarts at 1 for every Stone Name. For Blue Stones, plans 1-17 are main plans. For White Stones, plans 1-32 are main plans. Each main plan receives its own Yield % value.

Individual Yield % = Est. Weight / Rough Weight

The result is displayed in column R to two decimal places. Column H remains blank for main plans. A Blue Stone with 17 or fewer plans, or a White Stone with 32 or fewer plans, therefore contains only individual-yield rows.

15.6 Additional plans after the stone-type main-plan limit

For Blue Stones, every row after plan 17 is an additional plan. For White Stones, every row after plan 32 is an additional plan. Additional rows remain in source order and are divided into variable-sized groups by scanning Est. Weight from top to bottom:

- The first row after the applicable main-plan limit always starts the first additional group: plan 18 for Blue Stones and plan 33 for White Stones.
- Round the current and previous Est. Weight conventionally to two decimal places for grouping comparison only. If the rounded current value is less than or equal to the rounded previous value, it remains in the current group.
- A new additional group starts only when the current two-decimal comparison value is greater than the previous two-decimal comparison value. Third-decimal differences must not create a new group.
- The two-decimal increase is only a group boundary. It must never cause rows to be sorted or moved. The raw Est. Weight remains unchanged in the output and all Yield % calculations continue to use the original full-precision values.

Two-decimal grouping example: 0.024 ct rounds to 0.02 ct and 0.021 ct rounds to 0.02 ct. Therefore 0.024 is not treated as greater than 0.021 for additional-group detection, and no new group starts.

- The last additional group ends on the final row of that Stone Name.

15.7 Additional-group calculations and merged cells

Additional Group Yield % = Sum of Est. Weight for every row in the group / Rough Weight

- Column H is vertically merged across the group's rows and displays the count as N Pcs, for example 4 Pcs.
- Column R is vertically merged across the same rows and displays the combined group Yield % once.
- The group may contain any positive number of rows; no fixed group size is assumed.
- Main-plan and additional-group yields are independent calculations within the same Stone Name.

Plans	Est. Weight sequence	Detected result
18-21	1.00, 0.95, 0.45, 0.05	One group; H = 4 Pcs; R = sum of four weights / Rough Weight
22-25	1.05, 0.85, 0.30, 0.08	New group starts because 1.05 > 0.05; H = 4 Pcs; merged group

Plans	Est. Weight sequence	Detected result
		yield

15.8 Required validation

- Stone Name must contain the delimiters required to parse Kapan, Packet and Signer Name.
- Rough Weight must be numeric, positive and consistent for every row of the same Stone Name.
- Shape must be present and Est. Weight must be numeric and non-negative.
- Invalid rows must be reported before download; they must not be silently rearranged or corrected.
- Values after the fixed first 11 columns are ignored and reported as a warning.

15.9 Current boundary

The utility prepares and calculates the planning workbook. It does not yet populate planning labels, map plans against category shortages, or choose the final commercially preferred plan. Those decisions remain controlled future training items.

15.10 Shape normalization

Before the Shape value is written to output column I, the raw planning Shape is normalized using the approved Sarin-to-ecosystem mapping. The mapping compares trimmed Shape text without case sensitivity. The ID field in the supplied reference workbook is not used.

Special conditional rule: when the Sarin Shape is EMERALD 5STEP, use Asscher only when Ratio is within 1.00-1.03 inclusive, and use Emerald when Ratio is greater than or equal to 1.40. Ratio is input column 8. A missing, non-numeric, below-1.00, or greater-than-1.03-and-below-1.40 Ratio is outside the approved ranges. The utility must highlight the row as a validation issue and stop export until it is corrected.

If a Sarin Shape is not present in the approved mapping, preserve the original Shape in the output and report a validation warning. Shape normalization changes only the displayed ecosystem Shape; it does not change planning row order, Est. Weight, additional-group detection or Yield % calculations.

Sarin Shape	Ecosystem Shape written to output
ROUND	Round
OLD ROUND	Old European Brilliant
LIYO MQ	Marquise
MQ_21	Antique Marquise
LeoPear11	Pear
LeoOval22	Oval
EMERALD 5STEP	Asscher for Ratio 1.00-1.03; Emerald for Ratio >= 1.40; otherwise validation issue
KRISS 3-STEP	Radiant Modified
RAD4(1)	Radiant
SQ.BE.CUS	Square Cushion Brilliant

Sarin Shape	Ecosystem Shape written to output
SQ_CU	Square Cushion Modified
BE.CU.LONG	Cushion Brilliant
CU.LONG	Cushion Modified
SQ.ANTIK.CUS	Square Antique Cushion
ANTIK-CU-LONG	Antique Cushion
BEZEL PRINCESS	Princess
S.HEART	Heart
KRISS OVAL	KRISS OVAL
STEP MQ	Step Marquise
ANT-OVAL	Antique Oval
Moval	Moval
FABRIZIO	Febrizio
LOZENGES-2	Lozenge Step Cut
Kite_V2	Kite
TREGAL(1)	Triangle
CAD	Cadillacs
TAPERED-BG-LW	Trapper Baguette
TRAP50	Trapezoid
BE-TRA	BRILLANT TRAPEZOID
HALF_MOON	Moon Half
BAGUETTE-LW	Baguette

15.11 Polished-weight display precision

Every Est. Weight in output column J must display exactly three decimal places. Examples: 1 becomes 1.000, 0.02 becomes 0.020, and 0.024 remains 0.024. This is a display rule for polished weight. The underlying original numeric value is retained for Yield % calculations. Additional-group boundary detection remains a separate rule and continues to compare values conventionally rounded to two decimal places only.

15.12 Additional-group highlighting

Only additional-group rows receive option highlighting. Columns H through Q are filled for each group; column R is excluded from the option-group fill. Main-plan rows are not option-highlighted.

A fixed sequence of 10 light pastel shades is used and restarts for every Stone Name. The first additional group is light blue, the second is light orange, and subsequent groups follow the same approved fixed sequence. If more than 10 groups exist, the sequence repeats.

15.13 Yield ranking highlights

Within each Stone Name, compare all calculated yield options: every individual main-plan Yield % and every combined additional-group Yield %. Highlight only the Yield % cell in column R for the three highest yields, using three fixed rank-specific shades. Rank 1, Rank 2 and Rank 3 must remain visually distinct. Yield % displays to two decimal places.